



Javier Sempere Hernández

Data Scientist & ML Engineer

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Data Scientist and ML Engineer with a strong mathematical background (BSc Mathematical Engineering, MSc Computational Statistics) and 2+ years of industry experience building and deploying machine learning systems in production. Hands-on expertise in generative models, federated learning, and MLOps. Track record of leading technical projects as consultant and researcher, working directly with clients in healthcare, cybersecurity, and space. Experienced in privacy-sensitive applications and scalable AI within microservices architectures.

Spanish: Native | **English:** C1

WORK EXPERIENCE

ML Engineer — *GMV GmbH – Darmstadt, Germany*

May 2025 – Present

- Led as sole consultant the end-to-end delivery of a Google Cloud-based ETL and BI pipeline for DLP analysis at a major European bank, now in production monitoring 5,000+ endpoints. Implemented an AI agent that autonomously generates periodic security-status reports for the bank's security officers, reducing manual reporting workload. Managed direct client engagement throughout, adapting the solution to evolving requirements.
- Led technical proposal writing for AI research and engineering projects submitted to the European Space Agency (ESA).
- Designed and implemented a VAE-based synthetic tabular data generator, enabling compliant data sharing under GDPR constraints.
- Built an LLM-powered AI agent integrated into a microservices architecture via Dapr, improving system modularity and enabling seamless inter-service communication.
- Processed and interpreted Sentinel-1 InSAR (L2/L3) products and Sentinel-2 optical imagery in support of Earth Observation research and production projects.
- Currently developing an online anomaly detection and predictive maintenance module for telemetry data from the Celeste mission, including agentic AI capabilities with RAG to support satellite operators.

Data Scientist — *GMV – Madrid, Spain*

Jan 2024 – Apr 2025

- Main developer of a production-ready federated learning platform (Python, PyTorch, gRPC, European Data Space connectors) for distributed training across clinical nodes; supported clients in developing deep learning models for medical applications.
- Contributed to VALLE, an European Space Agency (ESA) NAVISP Element 1 project, implementing privacy-preserving PNT techniques via homomorphic encryption and SMPC in Zig and Python for encrypted GNSS signal processing.
- Led an anomaly-detection project for industrial cybersecurity environments in collaboration with a research team from Universidad de Castilla-La Mancha (UCLM).
- Developed and deployed discriminative and generative ML models in production.

EDUCATION

MSc in Computational Statistics — *Universidad Politécnica de Madrid*

2023 – 2024

Convex optimisation, neural networks, Monte Carlo simulation, time series, and probabilistic graphical models. 60 ECTS.

BSc in Mathematical Engineering — *Universidad Complutense de Madrid*

2019 – 2023

Advanced pure and applied mathematics. Community of Madrid Excellence Scholar. 240 ECTS.

BSc in Physics (*part-time*) — *Universidad Nacional de Educación a Distancia*

2025 – Ongoing

Studying alongside full-time work, driven by personal interest in physics.

PUBLICATIONS

Ongoing collaboration with a research team at Universidad Politécnica de Madrid and MIT Media Lab, using Sentinel-1/2 data to estimate seismic building exposure and study urban access in areas that often lack official cadastral records; further publications forthcoming. Some of that research is being recorded in my website.

- Ó. Alcarria Laserna, R. Sánchez Navarro, **J. Sempere**, P. Torrijos, J. C. Alfaro, J. M. Auñón, J. A. Gámez, J. M. Puerta. *Structure-Aware Unsupervised Anomaly Detection for Spacecraft Telemetry with Adaptive EVT Thresholding*. SPAICE 2026 (accepted, pre-print coming soon).
- M. Ureña, J. Rodríguez, **J. Sempere**, B. Moya, S. Rodríguez, F. Chinesta, B. González, M. Marchamalo. *Machine learning classification of structural systems using exclusively geospatial and remote sensing attributes: A case study in Santo Domingo, Dominican Republic*. ECEE 2026 (Full Paper). (*UPM & MIT Media Lab collaboration*)
- R. Hermosilla, **J. Sempere**, M. Ureña, B. Moya, S. Rodríguez, F. Chinesta, B. González, M. Marchamalo. *Functional Regression of Capacity Curves for Masonry Row-House Aggregates* (Pre-print coming soon). (*UPM & MIT Media Lab collaboration*)
- **J. Sempere**. *Aproximación de puntos vía curvas de Bézier*. *Bachelor's Thesis in Mathematical Engineering*. Universidad Complutense de Madrid, 2023. [PDF link](#).
- **J. Sempere**. *Privacy-Enhancing Machine Learning*. *Master's Thesis in Computational Statistics*. Universidad Politécnica de Madrid, 2024. [PDF link](#).

AWARDS & RECOGNITION

- Selected for the inaugural ESA Academy Cybersecurity Training Course (Apr 2024, ESA ESEC, Belgium).
- 1st place, Mathematical Modelling Competition UCM 2021–22 – automated defect detection in textile manufacturing using statistical inference (problem proposed by GMV; led to a job offer).
- Finalist, Spanish Programming Olympiad (2019) – Olympiad aimed at those under 18 years of age. Competitive programming problems solved in C++.